TEXORA · LECTURE SHEET

Laplace Transform

Definition and existence, the standard transforms and their properties — shifting, derivatives, integrals, convolution, the delta function — and the three-step solution of ODEs with initial conditions built in.

MATH-1201 · MATHEMATICS-II · TOPIC 3 · LEVEL 1 TERM 2

1. Definition and existence

For f(t) defined for t ≥ 0, the **Laplace transform** is

$$F(s) = L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

It exists when f is piecewise continuous on every finite interval and of **exponential order** (|f(t)| ≤ Me^{at} for large t) — the **sufficient conditions** of the syllabus; then the integral converges for s > a. (Sufficient, not necessary: t^{−1/2} has a transform despite blowing up at 0.) The transform turns functions of t into functions of s, calculus into algebra.

2. The standard table

F(T)	F(S)
1	1/s
t^n (n = 0, 1, 2, …)	n!/s^{n+1}
e^{at}	1/(s − a)
sin at, cos at	a/(s² + a²), s/(s² + a²)
sinh at, cosh at	a/(s² − a²), s/(s² − a²)
Unit step u(t − a)	e^{−as}/s
Dirac delta δ(t − a)	e^{−as} (and L{δ(t)} = 1)

Linearity is immediate from the integral: L{αf + βg} = αF + βG — the table plus linearity already covers most combinations.

3. The working properties

- **First shifting theorem:** $L\{e^{at}f(t)\} = F(s - a)$. (E.g. $L\{e^{2t} \sin 3t\} = 3/((s - 2)^2 + 9)$.)
- **Second shifting theorem:** $L\{f(t - a)u(t - a)\} = e^{-as}F(s)$ — a delay in t is a factor e^{-as} in s.
- **Multiplication by t:** $L\{t\, f(t)\} = -F'(s)$; generally $L\{t^n f\} = (-1)^n F^{(n)}(s)$. Hence $L\{t\, e^{at}\} = 1/(s - a)^2$ and $L\{t \cos at\} = (s^2 - a^2)/(s^2 + a^2)^2$.
- **Division by t:** $L\{f(t)/t\} = \int_s^\infty F(u)\, du$.
- **Transform of a derivative:** $L\{f'(t)\} = sF(s) - f(0)$; $L\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$. *This is the property the whole method exists for* — initial conditions enter automatically.
- **Transform of an integral:** $L\{\int_0^t f\} = F(s)/s$.
- **Derivatives/integrals of the transform itself** correspond to multiplication/division by t (above) — the "derivatives and integrals of transforms" of the syllabus.
- **Periodic functions:** if f has period T, $L\{f\} = [1/(1 - e^{-sT})] \int_0^T e^{-st}f(t)\, dt$ — transform one period, then divide.
- **Initial and final value theorems:** $f(0^+) = \lim_{s\to\infty} sF(s)$; $f(\infty) = \lim_{s\to 0} sF(s)$ (when the limit exists). Quick audits of any computed solution.

4. Inverse transform and convolution

Inversion is done by recognising table forms, using partial fractions and the shifting theorems in reverse. The **convolution** of f and g,

$$(f * g)(t) = \int_0^t f(u)\, g(t - u)\, du, \quad \text{satisfies} \quad L\{f * g\} = F(s)\, G(s)$$

— the **convolution theorem** — which inverts products that partial fractions cannot split.

> **WORKED EXAMPLES**
>
> **Partial fractions:** $L^{-1}\{1/(s^2(s + 2))\}$: $1/(s^2(s+2)) = -1/(4s) + 1/(2s^2) + 1/(4(s+2))$ → inverse = **$-1/4 + t/2 + e^{-2t}/4$**.
>
> **Convolution:** $t * e^t = \int_0^t u\, e^{t-u}\, du = e^t - t - 1$. Check by transforms: $(1/s^2)(1/(s-1))$ and $L\{e^t - t - 1\} = 1/(s-1) - 1/s^2 - 1/s = 1/(s^2(s-1))$ ✓.

5. Solving ODEs — the three steps

Step 1: transform the equation (derivatives become s-powers minus initial values). **Step 2:** solve the resulting algebra for Y(s). **Step 3:** invert.

> **WORKED EXAMPLES**

(a) $y'' + 4y = 0$, $y(0) = 1$, $y'(0) = 0$: $s^2Y - s + 4Y = 0 \to Y = s/(s^2 + 4) \to y = \mathbf{\cos 2t}$.
(b) $y' + 2y = e^{-2t}$, $y(0) = 1$: $(s + 2)Y = 1 + 1/(s + 2) \to Y = 1/(s + 2) + 1/(s + 2)^2 \to y = \mathbf{e^{-2t} + t\,e^{-2t}}$ — the resonant forcing answers with the t-factor, as Topic 2's failure case promised.
Audit (b) by the value theorems: $sY(s)$ at $s \to \infty$ gives $y(0) = 1$ ✓; at $s \to 0$ gives $y(\infty) = 0$ ✓.

6. Key points

- Laplace trades differentiation for multiplication by s, with initial conditions absorbed at step 1 — no separate "apply the conditions" stage.
- Shifts: e^{at} in t shifts s; a delay in t multiplies by e^{-as}.
- Convolution inverts products: $L\{f * g\} = F \cdot G$; δ is its identity ($f * \delta = f$, matching $L\{\delta\} = 1$).
- Value theorems audit endpoints of a solution without solving anything.
- The Dirac delta models an instantaneous impulse (a hammer blow, a voltage spike): zero duration, unit area, transform e^{-as} for a blow at $t = a$.

7. Practice

Answer in your own words; brief model answers follow.

1. Find $L\{3e^{2t} - 2\sin 3t + t^3\}$.
2. Solve $y'' + y = 1$, $y(0) = y'(0) = 0$, by Laplace transform.
3. Find $L^{-1}\{ (2s + 3)/(s^2 + 4s + 13) \}$.
4. Use convolution to invert $1/(s^2(s - 1))$ indirectly: identify f and g whose transforms are $1/s^2$ and $1/(s - 1)$, state their convolution, and give the inverse.
5. For $F(s) = (s + 3)/((s + 1)(s + 2))$, use the value theorems to find $f(0^+)$ and $f(\infty)$.
6. Solve $y'' - y = 0$, $y(0) = 0$, $y'(0) = 1$, and name the resulting function.

Answer key

1. $3/(s - 2) - 6/(s^2 + 9) + 6/s^4$ (using $L\{t^3\} = 3!/s^4 = 6/s^4$).
2. $s^2Y + Y = 1/s \to Y = 1/(s(s^2 + 1)) = 1/s - s/(s^2 + 1) \to y = 1 - \cos t$. Check: $y'' + y = \cos t + 1 - \cos t = 1$ ✓, and $y(0) = y'(0) = 0$ ✓.
3. Complete the square: $s^2 + 4s + 13 = (s + 2)^2 + 9$. Numerator $2s + 3 = 2(s + 2) - 1$. So the transform $= 2(s+2)/((s+2)^2 + 9) - (1/3)\cdot[3/((s+2)^2 + 9)] \to$ inverse $= e^{-2t}[2\cos 3t - (1/3)\sin 3t]$.
4. $f(t) = t$ (transform $1/s^2$) and $g(t) = e^t$ (transform $1/(s - 1)$). Their convolution (from §4, with the roles as computed there) is $t * e^t = e^t - t - 1$, and by the convolution theorem that is exactly

$L^{-1}\{1/(s^2(s-1))\}$.

5. $f(0^+) = \lim_{s\to\infty} s(s+3)/((s+1)(s+2)) = 1$ (ratio of leading terms). $f(\infty) = \lim_{s\to 0} s(s+3)/((s+1)(s+2)) = 0$ — the response is a sum of decaying exponentials, as the poles at -1, -2 promise.
6. $s^2Y - 1 - Y = 0 \to Y = 1/(s^2-1) \to y = \sinh t$. (Indeed $\sinh'' = \sinh$, $\sinh 0 = 0$, $\cosh 0 = 1$ ✓.)
